INCH-POUND

MS51972B
6 DEC 2022
SUPERSEDING
MS51972A
29 JUN 1969

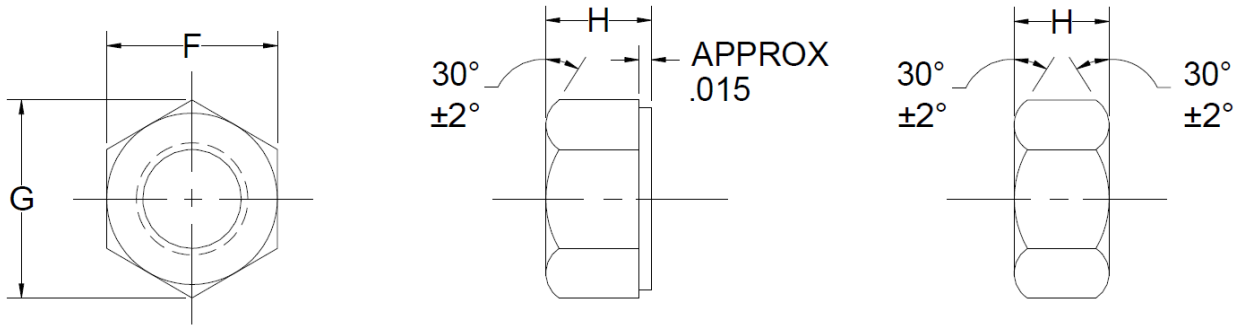
# DETAIL SPECIFICATION SHEET

## NUT: SQUARE, HEXAGON, CAP, SLOTTED, CASTLE, KNURLED, WELDING AND SINGLE BALL SEAT

Reactivated after 6 December 2022 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The Requirements for acquiring the product described herein shall consist of this specification sheet and FF-N-836.




ACCEPTABLE DESIGNS

Figure 1. NUT, PLAIN, HEXAGON

Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 6090 Strathmore Road, Richmond, VA 23297-5616 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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Table I. Dash Numbers and Dimensions

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NOMINAL SIZE OR BASIC MAJOR DIAMETER OF THREAD | | | 1/4 .2500 | 5/16 .3125 | 3/8 .3725 | 7/16 .4375 | 1/2 .5000 | 9/16 .5625 | 5/8 .6250 |
| THREADS PER INCH | | | 28 | 24 | 24 | 20 | 20 | 18 | 18 |
| F | WIDTH ACROSS FLATS | NOM | 7/16 | 1/2 | 9/16 | 11/16 | 3/4 | 7/8 | 15/16 |
| | | MAX | .4375 | .5000 | .5625 | .6875 | .7500 | .8750 | .9375 |
| | | MIN | .4280 | .4890 | .5510 | .6750 | .7360 | .8610 | .9220 |
| G | WIDTH ACROSS CORNERS | MAX | .505 | .577 | .650 | .794 | .866 | 1.010 | 1.083 |
| | | MIN | .488 | .557 | .628 | .768 | .840 | .982 | 1.051 |
| H | THICKNESS | NOM | 7/32 | 17/64 | 21/64 | 3/8 | 7/16 | 31/64 | 35/64 |
| | | MAX | .226 | .273 | .337 | .385 | .448 | .496 | .559 |
| | | MIN | .212 | .258 | .320 | .365 | .427 | .473 | .535 |
| MATERIAL | | | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. |
| STEEL, CORROSION RESISTING | | | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| | | | | | | | | | |
| NOMINAL SIZE OR BASIC MAJOR DIAMETER OF THREAD | | | 3/4 .7500 | 7/8 .8750 | 1 1.0000 | 1-1/8 1.1250 | 1-1/4 1.2500 | 1-3/8 1.3750 | 1-1/2 1.5000 |
| THREADS PER INCH | | | 16 | 14 | 12 | 12 | 12 | 12 | 12 |
| F | WIDTH ACROSS FLATS | NOM | 1-1/8 | 1-5/16 | 1-1/2 | 1-11/16 | 1-7/8 | 1-2/16 | 2-1/4 |
| | | MAX | 1.1250 | 1.3125 | 1.5000 | 1.6875 | 1.8750 | 2.0625 | 2.2500 |
| | | MIN | 1.0880 | 1.2690 | 1.4500 | 1.6310 | 1.8120 | 1.9940 | 2.1750 |
| G | WIDTH ACROSS CORNERS | MAX | 1.299 | 1.516 | 1.732 | 1.949 | 2.165 | 2.382 | 2.598 |
| | | MIN | 1.240 | 1.447 | 1.653 | 1.859 | 2.066 | 2.273 | 2.480 |
| H | THICKNESS | NOM | 41/64 | 3/4 | 55/64 | 31/32 | 1-1/16 | 1-11/64 | 1-9/32 |
| | | MAX | .665 | .776 | .887 | .999 | 1.094 | 1.206 | 1.317 |
| | | MIN | .617 | .724 | .831 | .939 | 1.030 | 1.138 | 1.245 |
| MATERIAL | | | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. |
| STEEL CORROSION RESISTING | | | 8 | 9 | 10 | 11 | 12 | 13 | 14 |

Requirements.

1. Material. Steel, corrosion resisting, 300 series in accordance with FF-N-836.

2. Protective coating. Passivate in accordance with SAE-AMS2700, Method 1 Type 6, 7, or 8, Class 4 or Method 2. Passivation treatments containing hexavalent chromium shall be prohibited.

3. Magnetic permeability. These fasteners have a magnetic permeability of 2.0 maximum (air=1.0) for a field strength of H=200 Oersteds using a magnetic indicator in accordance with ASTM A342/A342M, test method 1.

4. Threads. Threads shall be in accordance with FED-STD-H28/2 or ASME B1.1.



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5. Part number. The MS part number consists of the MS number plus the dash number. Example: MS51972 – 1

Notes.

1. Unless otherwise specified, all dimensions are in inches.

2. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. Referenced government (or non-government) documents of the issue listed in that issue of the department of defense index of specifications and standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

4. Do not use unassigned part numbers.

5. Changes from previous issue. Entire specification revised.

## INTERCHANGEABILITY TABLE

The following nuts covered by dash numbers given in MS35690 are cancelled after the date indicated on the cancellation document. Use the dash numbers given in the preceding sheet. The cancelled nuts cannot always replace the new nuts and should be used until existing stocks are depleted. Use only the new nuts for design and replacement. Replacement shall be in accordance with Table II. Interchangeability information regarding part numbers on the superseding drawing is also contained in this table.

Table II.

| CANCELED MS35690 DASH NO. | SUPERSEDING MS51972 DASH NO. | | CANCELED MS35690 DASH NO. | SUPERSEDING MS51972 DASH NO. | | CANCELED MS35690 DASH NO. | SUPERSEDING MS51972 DASH NO. |
|---|---|---|---|---|---|---|---|
| 430 | 1 | | 930 | 6 | | 1830 | 11 |
| 530 | 2 | | 1030 | 7 | | 2030 | 12 |
| 630 | 3 | | 1230 | 8 | | 2230 | 13 |
| 730 | 4 | | 1430 | 9 | | 2430 | 14 |
| 830 | 5 | | 1630 | 10 | | | |



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CONCLUDING MATERIAL

Custodians:
Army – AR
DLA – GS

Preparing Activity:
DLA – GS7

(Project 5310-2023-003)

Review Interests:
Army – AT, AV, MI
Navy – MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.


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